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Fibration Attention Dynamics

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Abstract

We characterize the dynamics of query updating coming from fibration abductive attention mechanisms, using perturbed Koopman operators that are invertible. Importantly, we identify conditions for Lyapunov stability of solutions near equilibrium points. Depending on the Koopman resolvent, some key-value pairings can have query update trajectories that are Lyapunov stable.

Synopsis

Fibration abduction attention mechanisms learn a fibration lifting neural network that simultaneously embodies query updating and abductive reasoning (finding keys that explain a value). Starting from an initial query, successive updates form a trajectory with unknown nonlinear dynamics, which we model by a Koopman operator. Koopman operators analyze nonlinear dynamical systems by transforming them into a linear framework where prediction is then straightforward. The trick is determining how to reverse this transformation, so as to make nonlinear predictions in the original space. We accomplish this by applying the representer theorem from reproducing kernel Hilbert space theory, to construct a vector valued Koopman operator whose components have perturbations that are invertible. So now the query trajectory can be predicted knowing its dynamics. It is then natural to inquire under what circumstances will a trajectory converge to an equilibrium point. Lyapunov stability analysis offers a solution provided one can find a Lyapunov function V ! Fortunately, for a Koopman operator K , a Lyapunov function V can be expressed in terms of resolvent of K . Moreover there are explicit conditions on V (a function of K) under which trajectory stability is achieved. The upshot is some key-value pairings can have fibration query update trajectories that converge near equilibrium points.

Fibration Attention Dynamics

Introduction

Tokens representing words are projected onto a lower dimensional latent space $\mathcal{X} \subseteq \mathbb{R}^d$. In $\mathcal{X}$, let subsets Q , K and V denote queries, keys and values respectively, and let Q' be a copy of Q , denote updated queries.

Data to learn neural networks from:

$$q_i \in Q, \quad i \in [n] = \{1, 2, \dots, n\}$$

$$(k_j, v_j) \in K \times V, \quad j \in [m]$$

+

Let $L: \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}^+$ be a loss function.

For $\varepsilon > 0$, define $x \approx_\varepsilon y$ iff $L(x, y) < \varepsilon$.

Suppose there exists $\varepsilon > 0$ such that we can learn a neural network $p: K \rightarrow V$ with $p(k_j) \approx_\varepsilon v_j, j \in [m]$,

we will use such a key-value semantic pairing to construct a weak factorization system fibration.

Transformer attention via fibration [1]

In a category with neural network arrows, consider the following Joyal-Tierney symbolic calculus diagram:

$$\begin{array}{ccc}
 Q \otimes K & \xrightarrow{m \otimes 1} & Q' \otimes K \\
 \downarrow 1 \otimes p & \nearrow \sigma & \downarrow 1 \otimes p \\
 Q \otimes V & \xrightarrow{m \otimes 1} & Q' \otimes V
 \end{array} \quad (+)$$

Learn lifting neural network $\sigma: Q \otimes V \rightarrow Q' \otimes K$

so that (+) commutes up to $=_\varepsilon$, that is:

$$i) \quad \sigma \begin{bmatrix} q_i \\ p(k_j) \end{bmatrix} =_\varepsilon \begin{bmatrix} m(q_i) \\ k_j \end{bmatrix} \quad i \in [n], j \in [m]$$

$$ii) \quad \pi_1 \sigma \begin{bmatrix} q_i \\ v_j \end{bmatrix} =_\varepsilon m(q_i) \quad i \in [n], j \in [m]$$

$$\text{iii)} \quad p \pi_2 \sigma \begin{bmatrix} q_i \\ v_j \end{bmatrix} =_\varepsilon v_j \quad i \in [n], j \in [m]$$

+

Query update step from q_i to $\vec{q}_i$

Let $q'_i \triangleq \sum_{j=1}^m \alpha_{ij} v_j$, where $\alpha_{ij} \triangleq \text{softmax score}(q_i, k_j)$,

score a similarity function eg a reproducing kernel,

and let $j^* = \arg \max_j \alpha_{ij}$.

In generic transformer attention, a feedforward neural

network is then applied to update q'_i , and such updates

successively iterated until perplexity is acceptable.

Fibration attention on the other hand learns a

fibration lifting neural network $\sigma = m \otimes \tau$, where

$p \circ \tau = 1_V$, to apply to q'_i .

Fibration attention update of q_i is defined to be

$$\vec{q}_i \triangleq \pi, \sigma \begin{bmatrix} q'_i \\ v_{j^*} \end{bmatrix}, \text{ which takes into account}$$

the key-value semantics of $p: K \rightarrow V$.

✗

Query update trajectories [2]

For each position $i \in [n]$,

$$\text{let } q_{i,0} = q_i,$$

$$q_{i,1} = \vec{q}_{i,0},$$

$$\vdots$$

$$q_{i,T} = q_{i,T-1},$$

be a trajectory in $\mathcal{X} \subseteq \mathbb{R}^d$ of query updates.

We model such trajectories by a Koopman operator

to predict $q_{i,T+1}$ and so on, in order to understand

the nonlinear dynamics of query updates From Fibration
abductive attention.

Let $k_2 : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ be any kernel,

the reproducing kernel $k(x, y) \triangleq \sum_{p=1}^d x_p y_p + k_2(x, y)$

satisfies $\pi_p : \mathbb{R}^d \rightarrow \mathbb{R} \in H(k)$, $\forall p \in [d]$
 $(z_1, \dots, z_d) \mapsto z_p$

and $H(k_2) \subseteq H(k)$. So without loss of generality

let $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ be a reproducing kernel with

$\pi_p \in H(k)$, $p \in [d]$.

For each key k_j , $j \in [m]$ define observable map

$g_j \in H(k) \subseteq \text{Fun}(\mathcal{X}, \mathbb{R})$ by $g_j \triangleq k(k_j, \cdot)$.

For each position $i \in [n]$, we have Koopman data:

a) query update trajectory $q_{i,0}, \dots, q_{i,T} \in \mathbb{R}^d$

b) observable maps $g_1, \dots, g_m \in H(k)$, $g_j \triangleq k(k_j, \cdot)$

c) measurements $y_{j,t} \triangleq g_j(q_{i,t}) = k(k_j, q_{i,t})$

for $j \in [m]$, $t = 0, 1, \dots, T$.

Following [3]:

- ① Determine vector-valued Koopman operator

$$\hat{K} \triangleq \bigoplus_{p \in [d]} \hat{K}_p : H^d \rightarrow H^d$$

From the constrained Representer Theorems

for $\hat{K}_p : H \rightarrow H$ and the query update trajectory.

- ② Determine Koopman embedding $\varphi \in H^d$,

$$\varphi = [\varphi_p] : \mathcal{X} \rightarrow \mathbb{R}^d, \quad \varphi_p \triangleq \hat{K}_p \pi_p, \quad p \in [d]$$

- ③ To proxy φ , draw* invertible perturbation

$$\phi \triangleq [\mathcal{K}_p \pi_p]$$

- ④ Define Koopman prediction by

$$\hat{q}_{i,T+1} \triangleq \phi^{-1}([\hat{K} \varphi](q_{i,T}))$$

* Perturbation ϕ is a function of two hyperparameters, $\lambda > 0$ and $\varepsilon > 0$; draw them from distribution on $(0,1]$ concentrated at 0. Then the nonlinear Koopman predictions are stochastic.

Lyapunov stability analysis

For the dynamical system (dependent on $p: K \rightarrow V$)

$$q_{i,t+1} = \vec{q}_{i,t}$$

we determined Koopman operators $\hat{K}_p$, $p \in [d]$

Let $r \in \mathbb{Z}^+$, define Lyapunov functions, for $p \in [d]$,

$$\begin{aligned} V_p: \mathcal{X} &\longrightarrow \mathbb{R}^+ \\ x &\longmapsto [(I - \hat{K}_p)^{-1} f](x) \end{aligned}$$

where $(I - \hat{K}_p)^{-1}$ is a Koopman resolvent,

$$\begin{aligned} \text{and } f: \mathcal{X} &\longrightarrow \mathbb{R} \\ y &\longmapsto \|y\|^r \end{aligned}$$

Theorem [4]

The equilibrium solution, $x = 0$, is r^{th} moment exponentially stable iff there exist positive constants

a, b, c with $c < 1$, such that

$$a \|x\|^r \leq V_p(x) \leq b \|x\|^r$$

$$\text{and } [\hat{K}_p V_p](x) \leq c V_p(x).$$

—

This result establishes conditions under which fibration attention query update trajectories converge near equilibrium points.

□

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